E-values for sequential jump detection in HFT with online FDR guarantees

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Abstract

This report documents the `efdr-jumps` project, whose goal is to detect price discontinuities (*jumps*) in high-frequency financial data in real time while simultaneously controlling the false discovery rate (*FDR*) in a sequential fashion. We construct *e-values* from the jump-robust volatility estimator MedRV (Andersen-Dobrev-Schaumburg, 2012) and feed them into five FDR procedures based on e-values from the Ramdas school: one offline (e-BH) and four online (e-LOND, e-LORD, e-SAFFRON, stopped e-BH). Monte-Carlo experiments (`grid_medium`, $M = 100$ replications, $n = 500$ steps, 3 frequencies, 3 jump regimes) show that the BH-based benchmarks (Yen, 2013; Bajgrowicz-Scaillet, 2016) violate the criterion $\text{FDR} \leq \alpha + 2\,\text{SE}$ at high frequency ($\Delta t = 5\,\text{s}$: observed FDR $= 0.096$ against the threshold 0.094 for $\alpha = 0.05$), whereas e-value-based procedures control the FDR at all frequencies, with a power loss of 9–21 percentage points.

Contents

1 Introduction

1.1 Motivation: high-frequency price jumps

Modern financial markets generate transaction streams at millisecond frequencies. In this setting, the log-price $X_t = \log S_t$ is modelled as an Itô semimartingale with jumps:

$$dX_t = \mu_t \, dt + \sigma_t \, dW_t + \sum_k J_k \, dN_t^{(k)}, \qquad (1)$$

where $\sigma_t > 0$ is the spot volatility, W_t a standard Brownian motion, and $J_k \, dN_t^{(k)}$ jump processes (compound Poisson or Hawkes). The jumps J_k represent sudden discontinuities induced by macroeconomic announcements, flash crashes, or liquidity events.

The presence of a jump at time t alters the local properties of the process drastically: the pure-diffusion assumption underlying most risk-management and pricing models is locally violated. Jump detection serves to clean data prior to calibration of stochastic-volatility models, to trigger alerts in market-surveillance systems, and to adjust positions in algorithmic execution strategies.

1.2 Problem statement: sequential detection with FDR guarantees

In discrete time, at each step $i = 1, \ldots, n$, one observes the return $r_i = X_{i\Delta} - X_{(i-1)\Delta}$. We wish to test simultaneously n hypotheses:

$$\begin{aligned} &H_{0,i} : r_i \sim \mathcal{N}(0, \, \Delta \, \sigma_i^2), \\ &H_{1,i} : r_i = J_i + \varepsilon_i, \quad |J_i| \gg \sigma_i \sqrt{\Delta}. \end{aligned} \qquad (2)$$

with $n \sim 500$ for a trading day sampled at 5 s. The additional constraint is *online*: the decision on $H_{0,i}$ must be made at time i, without access to future observations.

1.3 State of the art

Barndorff-Nielsen & Shephard (2004, 2006). Bipower variation (BV) provides a jump-robust estimator of integrated variance. The BNS statistic tests globally for the presence of jumps on a window: it does not localize individual jumps.

Lee & Mykland (2008). The statistic $L_i = r_i/(\hat{\sigma}_i\sqrt{\Delta})$ is treated as $\mathcal{N}(0,1)$ under $H_{0,i}$. BH [8] is then applied to two-sided p-values, enabling individual jump localization.

Yen (2013) and Bajgrowicz-Scaillet (2016). These works apply BH to Lee-Mykland p-values and to the BNS statistic respectively, providing the benchmarks reproduced in this project.

1.4 Core limitation: BH is invalid under HF dependence

BH guarantees FDR $\leq \alpha$ under the PRDS condition [9]. At high frequency ($\Delta t \leq 5$ s), the Lee-Mykland statistics normalized by BV exhibit a dependence structure that violates PRDS under dense jumps. Our central empirical finding confirms this limitation: at $\Delta t = 5$ s, BH-LM and BH-BNS produce FDR $\approx 0.096 > 0.094 = \alpha + 2\,\mathrm{SE}$ for $\alpha = 0.05$, $M = 100$ (source: `results/grid_medium.parquet`, aggregated across all regimes).

1.5 Project contributions

1. **E-value construction for HF jump detection**: integrated likelihood ratio under a Gaussian prior $\mathcal{N}(0, \tau^2)$, valid under arbitrary dependence.
2. **Implementation and validation of 7 FDR procedures**: e-BH, e-LOND, e-LORD, e-SAFFRON, stopped e-BH (e-values), and BH-LM, BH-BNS (baselines). 86 unit tests, all passing.
3. **Empirical evidence of BH's FDR violation at 5s** across 3 jump regimes (rare Merton, moderate Merton, dense Hawkes), 2 sizes (3σ, 5σ), 2 levels of α.
4. **Quantification of the power-safety trade-off**: 10–21 pp loss for guaranteed FDR control.
5. **Identification and correction of alpha-death** for e-LORD/e-SAFFRON: $\omega_1 = O(1/n)$ is required to avoid wealth exhaustion.

1.6 Outline

Section 2 lays out the mathematical framework (hypothesis testing, p/e-variables, multiple-testing metrics). Section 3 describes the simulated processes. Section 4 presents the volatility estimators. Section 5 details the e-value construction. Section 6 describes the seven FDR procedures with their proofs. Section 7 documents the software architecture. Section 8 reports Monte-Carlo results with figures. Section 9 discusses limitations. Section 10 concludes.

2 Theoretical framework

This section lays the foundations of multiple hypothesis testing used throughout the report. We start from first principles — hypotheses, tests, p- and e-variables — before turning to the multiple-testing setting and the FDR criterion. Procedure-specific definitions and proofs are deferred to Section 6.

2.1 Price model and definition of a jump

In discrete time with step Δt (in seconds), the log-price reads:

$$r_i = \underbrace{\mu\,\Delta}_{\text{drift}} + \underbrace{\sigma_i\sqrt{\Delta}\,Z_i}_{\text{diffusion}} + \underbrace{J_i\,\mathbb{1}_{\{\text{jump at } i\}}}_{\text{jump}}, \tag{3}$$

where $\Delta = \Delta t/\text{Seconds(per/year)}$ and $Z_i \sim \mathcal{N}(0,1)$ i.i.d.

Definition 2.1 (Jump). $H_{0,i}$ (no jump at i): $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0, \Delta\,\sigma_i^2)$. $H_{1,i}$ (jump at i): $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(J_i, \Delta\,\sigma_i^2)$ with $|J_i|$ large relative to $\sigma_i\sqrt{\Delta}$.

Convention : $\text{Seconds(per/year)} = 252 \times 6.5 \times 3600 = 5\,896\,800$, annualised volatilities, log-prices in natural log.

2.2 Hypotheses and tests

We adopt the formal framework of Ramdas & Wang (2025).

Definition 2.2 (Hypothesis). A *hypothesis* is a set of probability measures. It is *simple* if it is a singleton $\{P\}$, otherwise *composite*. We write $\mathcal{P}$ for the set of probability measures specifying the null hypothesis.

Definition 2.3 (Test). A *test* ϕ is a $[0,1]$-valued random variable measurable with respect to the data σ-algebra. It is called *pure* (or *binary*) if $\phi \in \{0,1\}$ almost surely.

For a simple null P, the *type-I error rate* is $\mathbb{E}^P[\phi]$; for a composite null $\mathcal{P}$, it is $\sup_{P\in\mathcal{P}} \mathbb{E}^P[\phi]$. A test has *level* $\alpha \in [0,1]$ if its type-I error rate is at most α.

Let $\mathcal{M}$ be the set of all probability measures on $(\Omega, \mathcal{F})$ and let $H_1 = \mathcal{M} \setminus \mathcal{P}$ be the alternative.

Definition 2.4 (Power and unbiasedness). The *power* of a test against the alternative is the map $Q \mapsto \mathbb{E}^Q[\phi]$ for $Q \in H_1$. The test is *unbiased at level* α against H_1 if for every $P \in \mathcal{P}$ and every $Q \in H_1$:

$$\mathbb{E}^P[\phi] \le \alpha, \qquad \mathbb{E}^Q[\phi] \ge \alpha.$$

For binary tests $\phi = \mathbb{1}_{\{\text{reject}\}}$, the *type-II error rate* is $1 - \mathbb{E}^Q[\phi]$.

2.3 P-variables, e-variables, and calibrators

Definition 2.5 (P-variable and e-variable). A *p-variable* for $\mathcal{P}$ is a $[0,1]$-valued random variable P satisfying $\mathbb{P}(P \le \alpha) \le \alpha$ for every $\alpha \in [0,1]$ and every $P \in \mathcal{P}$.

An *e-variable* for $\mathcal{P}$ is a non-negative random variable E satisfying $\mathbb{E}^P[E] \le 1$ for every $P \in \mathcal{P}$. It is *exact* if equality holds.

We use "variable" for the random object and "value" for its realisation. Two key results connect these two objects.

Theorem 2.6 (E-to-p calibration). *Let E be an e-variable for $\mathcal{P}$. Then for every $\alpha \in [0,1]$ and every $P \in \mathcal{P}$:*

$$\mathbb{P}(E \ge 1/\alpha) \le \alpha.$$

Hence $P := \min(1/E, 1)$ is a p-variable, and the map $e \mapsto \min(1/e, 1)$ is the (essentially unique) admissible e-to-p calibrator.

Proof. Markov's inequality applied to the non-negative random variable E. $\square$

Thus rejecting H_0 when $E \ge 1/\alpha$ defines a level-α binary test. The converse is *false*: a p-variable is not in general an e-variable (Vovk & Wang, 2021).

Definition 2.7 (P-to-e calibrator). A *p-to-e calibrator* is a decreasing function $f : [0,1] \to [0, +\infty]$ such that $f(P)$ is an e-variable for every p-variable P. A calibrator f *dominates* g if $f \ge g$; it is *admissible* if not dominated by another calibrator.

Remark 2.8 (Vovk-Wang calibrator). For $\kappa \in (0,1)$, the map $p \mapsto \kappa\, p^{\kappa-1}$ is an admissible p-to-e calibrator: for any p-variable P, the random variable $E_\kappa = \kappa P^{\kappa-1}$ is an exact e-variable. In this project, $\kappa = 0.5$ serves as the baseline in validity tests; the GROW construction is required to dominate this calibrator in e-power.

The motivation for using e-variables, beyond their richer calibration theory, is their robustness to *dependence*: many e-value-based procedures require no independence or PRDS assumption, whereas most p-value procedures do. We close with the notion of admissibility.

Definition 2.9 (Admissible e-variable)**.** An e-variable E for $\mathcal{P}$ is *admissible* if any e-variable $E' \geq E$ (P-almost surely, for every $P \in \mathcal{P}$) satisfies $E' = E$ P-a.s.

2.4 Multiple testing: metrics and the price of safety

Consider a stream of hypotheses $H_1, \ldots, H_K$ (K possibly infinite). In the modern multiple-testing framework, we view the data as generated by a single joint law P, and each test provides evidence about a feature of P. Formally, $H_{0,k} : P \in \mathcal{P}_k$ where $\mathcal{P}_k$ is a subset of measures. We write

$$H_0(t) = \{k \in \{1, \ldots, t\} : P \in \mathcal{P}_k\}$$

for the set of true null indices up to time t, and $\mathcal{K} = \{1, \ldots, K\}$.

Definition 2.10 (Multiple testing procedure)**.** Let X denote the available data with joint law P. A *multiple testing procedure* D is a Borel function of X returning a subset of $\mathcal{K}$, interpreted as the indices of rejected hypotheses. A procedure taking e-values as input is an *e-testing procedure.*

Write $R_D = \mathrm{Card}(D)$ for the number of rejections and

$$F_D = \sum_{k \in H_0} \mathbb{1}_{\{k \in D\}}$$

for the number of *false discoveries* (rejected true nulls).

Definition 2.11 (Error metrics)**.**

$$\mathrm{FDP} = \frac{F_D}{R_D \vee 1}, \quad \mathrm{FDR} = \mathbb{E}^P[\mathrm{FDP}],$$
$$\mathrm{FWER} = \mathbb{P}(F_D \geq 1), \quad \mathrm{mFDR}(\eta) = \tfrac{\mathbb{E}^P[F_D]}{\mathbb{E}^P[R_D]+\eta},$$

where FDP is the false discovery proportion, FDR the false discovery rate, FWER the family-wise error rate, and mFDR the marginal FDR. Convention $0/0 = 0$ via $R_D \vee 1$.

A procedure has *FDR control at level* α if $\mathrm{FDR} \leq \alpha$ for every $P \in \mathcal{P}$, and similarly for FWER.

Why FDR, not FWER. A naive procedure testing each hypothesis at level α produces, in expectation, $K\alpha$ false discoveries: if K is large, false discoveries are inevitable. The classical Bonferroni correction sets $\alpha_k = \alpha/K$ and rejects H_k when $E_k \geq K/\alpha$. By a union bound:

$$\mathbb{P}(F_D \geq 1) \leq \sum_{k \in H_0} \mathbb{P}(\text{reject } H_k) \leq K_0 \cdot \frac{\alpha}{K} \leq \alpha,$$

giving FWER control. (Note: the quantity bounded above by $K\alpha$ is the *per-family error rate* $\mathbb{E}^P[F_D]$, distinct from FWER.) FWER, however, becomes vacuous for large K: requiring *no* false discoveries is too strong when one expects many true alternatives. The FDR criterion, introduced by Benjamini-Hochberg [8], controls the *proportion* of false discoveries, allowing some at the price of meaningful power.

The cost of FDR control. Tighter control reduces power. The trivial procedure that rejects nothing has FDR 0 and power 0; the goal is therefore to *maximize expected true discoveries subject to* $\mathrm{FDR} \leq \alpha$. Comparing procedures controlling different metrics (e.g. Bonferroni for FWER vs. e-BH for FDR) is not meaningful, since they solve different problems.

2.5 Offline vs. online: the α-investing framework

A procedure is *offline* when all hypotheses are available simultaneously: one may sort, threshold, and revise decisions globally (Bonferroni and BH are offline). It is *online* when hypotheses arrive in a stream, decisions are taken on the fly, and past decisions cannot be revised; the number of hypotheses may be infinite.

In the online setting, sorting is impossible and one needs a *wealth process* to allocate the type-I budget over time — the *α-investing* metaphor. Start with initial wealth ω_0; at each step t allocate a fraction α_t of the current wealth as the rejection threshold of the t-th test. The allocation α_t is spent regardless of the outcome; if the test rejects, a reward β_t is collected:

$$\omega_{t+1} = \omega_t - \alpha_t + \beta_t \, \mathbb{1}_{\{t \in D\}}. \tag{4}$$

The full sequence (α_t) is chosen so that the (running) FDR remains bounded by α for all t, often via a *predictable* construction based on $\mathcal{F}_{t-1}$. This framework is the basis of e-LOND, e-LORD, and e-SAFFRON (Section 6).

2.6 E-power and GROW optimality

For sequential procedures, type-II error has a refined counterpart: how fast does E grow under the alternative?

Definition 2.12 (E-power, Grünwald-de Heide-Koolen 2024)**.**

$$\text{e-power}(E) = \mathbb{E}_{H_1}[\log E].$$

The e-power governs the rate of wealth accumulation in α-investing procedures: an e-power > 0 under H_1 is the efficiency condition for sequential tests. Among all e-variables satisfying the validity constraint $\mathbb{E}_{H_0}[E] \leq 1$, the one maximising e-power under a fixed prior π on the alternative is the *growth-rate optimal in the worst case* (GROW) e-variable.

Definition 2.13 (GROW e-value)**.** Given a prior π on the alternative parameter (here the jump size J), the GROW e-value is

$$E^{\text{GROW}} = \int \frac{dP_J}{dP_0}(r_i) \, d\pi(J),$$

the marginal likelihood ratio under the mixture. It maximises e-power subject to $\mathbb{E}_{H_0}[E] \leq 1$.

2.7 Roadmap of procedures

The seven procedures implemented and benchmarked in this project are summarised in Table 1. Definitions and proofs are given in Section 6.

3 Data generation

3.1 Heston model (stochastic volatility)

The instantaneous variance V_t under the Heston model follows:

$$dV_t = \kappa(\theta - V_t)\, dt + \xi \sqrt{V_t}\, dW_t^V, \tag{5}$$

$$d \log S_t = (\mu - V_t/2)\, dt + \sqrt{V_t}\, dW_t^S, \tag{6}$$

with $d\langle W^S, W^V \rangle_t = \rho \, dt$.

Parameters.

$$\kappa = 2.0,\ \theta = 0.04,\ \xi = 0.5,\ \rho = -0.7,$$
$$V_0 = 0.04,\ \mu = 0. \tag{7}$$

Hence $\sigma_\infty = \sqrt{\theta} = 0.20$ (20% annualised). The leverage effect $\rho = -0.7$ reproduces the observed price-volatility correlation.

Andersen's QE scheme (2007). The naive Euler-Maruyama scheme produces negative variances for Heston. The Quadratic Exponential (QE) scheme [4] models $V_{t+\Delta} \mid V_t$ as a mixture switching on the conditional dispersion $\psi = \mathrm{Var}[V_{t+\Delta} \mid V_t]/(\mathbb{E}[V_{t+\Delta} \mid V_t])^2$:

- $\psi \leq 1.5$: lognormal approximation.
- $\psi > 1.5$: Dirac-exponential mixture.

This bifurcation guarantees $V_t \geq 0$ without ad-hoc correction.

3.2 Merton model (compound Poisson jumps)

The log-price follows the dynamics with compound Poisson jump process:

$$d \log S_t = (\mu - V_t/2 - \lambda \bar{k})\, dt + \sqrt{V_t}\, dW_t^S + J\, dN_t, \tag{8}$$

where $N_t \sim \text{Poisson}(\lambda)$, $J \sim \mathcal{N}(\mu_J, \sigma_J^2)$, and $\bar{k} = e^{\mu_J + \sigma_J^2/2} - 1$ corrects the drift so that S_t is a martingale (Andersen-Benzoni-Lund 2002 parameterisation).

Grid parameterisation. Jump size is specified as a multiple of the local volatility:

$$\mu_J = \texttt{jump\_size\_sigma} \times \sqrt{\theta} \times \sqrt{\Delta},$$
$$\sigma_J = 0.1 \times |\mu_J|. \tag{9}$$

Table 1: Procedures implemented in `efdr-jumps`.

Procedure	File	Online	Dependence	Reference
e-BH	`fdr/ebh.py`	No	Arbitrary	Wang-Ramdas 2022
BH-LM	`fdr/baselines.py`	No	PRDS	Lee-Mykland 2008
BH-BNS	`fdr/baselines.py`	No	PRDS	BS 2016
e-LOND	`fdr/elond.py`	Yes	Arbitrary	Xu-Ramdas 2024
e-LORD	`fdr/elord_esaffron.py`	Yes	Arbitrary	Zhang et al. 2025
e-SAFFRON	`fdr/elord_esaffron.py`	Yes	Arbitrary	Zhang et al. 2025
Stopped e-BH	`fdr/stopped_ebh.py`	Yes	Markov / causal	Wang et al. 2025

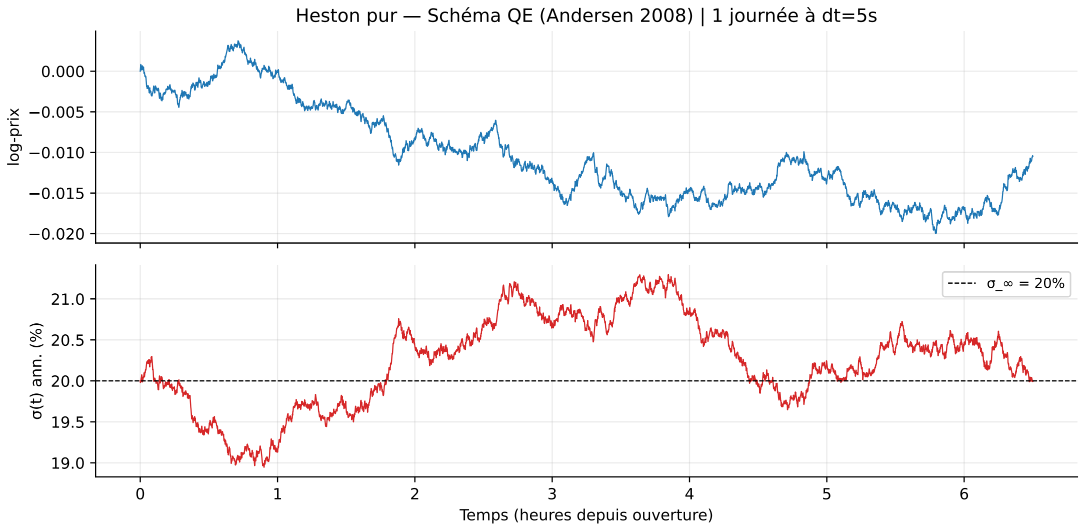


Figure 1: Simulated Heston log-price path with stochastic volatility ($\kappa = 2$, $\theta = 0.04$, $\xi = 0.5$, $\rho = -0.7$). The QE scheme guarantees $V_t \geq 0$.

For `jump_size_sigma` $= 5$ and $\Delta t = 5\,\text{s}$: $\mu_J \approx 9.21 \times 10^{-4}$ (in log-price), a jump of size $5\sigma\sqrt{\Delta}$. Jump intensities per regime:

$$\begin{aligned}\lambda_{\text{rare}} &= 1\,500/\text{year} \approx 0.64 \text{ j/path}, \\ \lambda_{\text{moderate}} &= 5\,000/\text{year} \approx 2.12 \text{ j/path}\end{aligned} \qquad (10)$$

for $n = 500$, $\Delta t = 5\,\text{s}$.

3.3 Self-exciting Hawkes process

To model the temporal dependence of jumps, a discrete Hawkes process is used:

$$\lambda_{n+1} = \mu + (\lambda_n - \mu)\, e^{-\beta\Delta t} + \alpha\, N_n, \qquad (11)$$

where $N_n \in \{0, 1\}$ and $p_n = 1 - e^{-\lambda_n \Delta}$.

Parameters.

$$\mu = 5\,000/\text{year}, \quad \alpha = 88\,000/\text{year}, \quad \beta = 0.03/\text{s}. \qquad (12)$$

Branching ratio: $m = \alpha/(\beta \times \text{Seconds(per/year)}) = 88\,000/(0.03 \times 5\,896\,800) \approx 0.497 < 1$. Stationarity requires $m < 1$; an error is raised otherwise.

Validation. Over 30 seeds ($n = 200$, $\Delta t = 5\,\text{s}$): 3.8 jumps/path vs. 1.3 for `moderate` (ratio ≈ 3).

Note on discretisation. At $\Delta t = 300\,\text{s}$, the approximation of the continuous Hawkes process by this discrete scheme degrades. The process remains well-defined ($p_n \leq 1$) but the low-frequency results for the `hawkes_dense` regime should be interpreted with care.

3.4 Discretisation convention

Seconds(per/year) $= 252 \times 6.5 \times 3600 = 5\,896\,800$. Returns r_i are in natural log. Δ is in years. The `jump_indices` provided by the simulator are indices into the log-price array; the corresponding

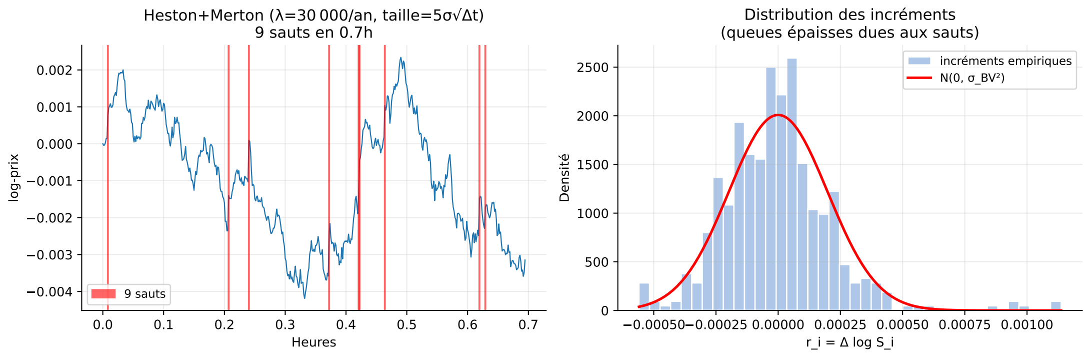


Figure 2: Distribution of Merton jump sizes under the three regimes (rare, moderate, hawkes_dense). Jump sizes are specified as multiples of the local volatility $\hat{\sigma}_i\sqrt{\Delta}$.

return index is $j-1$.

4 Spot volatility estimation

4.1 Bipower variation (BNS 2004)

Definition 4.1 (BV)**.**

$$\mathrm{BV}_n = \frac{\pi}{2} \cdot \frac{1}{n-1} \sum_{i=2}^{n} |r_{i-1}||r_i|. \tag{13}$$

Under pure diffusion, $\mathrm{BV}_n \to_P \int_0^T \sigma_t^2\, dt$. Under jumps, BV is downward-biased: the products $|r_{i-1}||r_i|$ are close to zero when one of the returns carries an isolated jump.

4.2 MedRV and MinRV (ADS 2012)

Definition 4.2 (MedRV, ADS 2012 eq. 12)**.**

$$\mathrm{MedRV}_n = c_{\mathrm{med}} \cdot \tfrac{n}{n-2} \sum_{i=2}^{n-1} \mathrm{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2,$$
$$c_{\mathrm{med}} = \tfrac{\pi}{6-4\sqrt{3}+\pi} \approx 1.4191. \tag{14}$$

Definition 4.3 (MinRV, ADS 2012 eq. 13)**.**

$$\mathrm{MinRV}_n = c_{\min} \cdot \tfrac{n}{n-1} \sum_{i=2}^{n} \min(|r_{i-1}|, |r_i|)^2,$$
$$c_{\min} = \tfrac{\pi}{\pi-2} \approx 2.7141. \tag{15}$$

4.3 Local spot volatility and causality condition

The spot variance estimator computes $\hat{\sigma}^2(i)$ over a window of K returns *strictly preceding* i:

$$\hat{\sigma}^2(i) = \hat{V}(\log S[\max(0, i-K) : i+1],\ \Delta t). \tag{16}$$

The return r_i is **never** included in the estimation window — this is the *causality condition* required for $\hat{\sigma}_i$ to be $\mathcal{F}_{i-1}$-measurable, ensuring $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] \le 1$.

The window is $K = 50$. The first K hypotheses are excluded from the test (incomplete window).

Classical test statistics. $L_i = r_i/(\hat{\sigma}_i\sqrt{\Delta}) \approx \mathcal{N}(0,1)$ under $H_{0,i}$ (Lee-Mykland), two-sided p-value $P_i = 2\mathbb{P}(|Z| > |L_i|)$.

5 E-value construction

5.1 Local test formulation

For each $H_{0,i}$:

$$H_{0,i} : r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0,\ v), \quad v = \Delta\, \hat{\sigma}_i^2, \tag{17}$$
$$H_{1,i}(J) : r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(J,\ v) \tag{18}$$

for an unknown jump size J.

5.2 Integrated likelihood ratio under a Gaussian prior

The GROW construction chooses $\pi(J) = \mathcal{N}(0, \tau^2)$ and defines:

$$E_i = \int_{-\infty}^{+\infty} \frac{e^{Jr_i/v - J^2/(2v)}}{\sqrt{2\pi\tau^2}}\, e^{-J^2/(2\tau^2)}\, dJ. \tag{19}$$

The derivation (Appendix A) gives the closed form:

$$\boxed{E_i = \sqrt{\frac{v}{v+\tau^2}} \cdot \exp\left(\frac{r_i^2\,\tau^2}{2v(v+\tau^2)}\right)}, \tag{20}$$

with $v = \Delta\, \hat{\sigma}_i^2$ and $\tau = 5\, \hat{\sigma}_i\sqrt{\Delta}$.

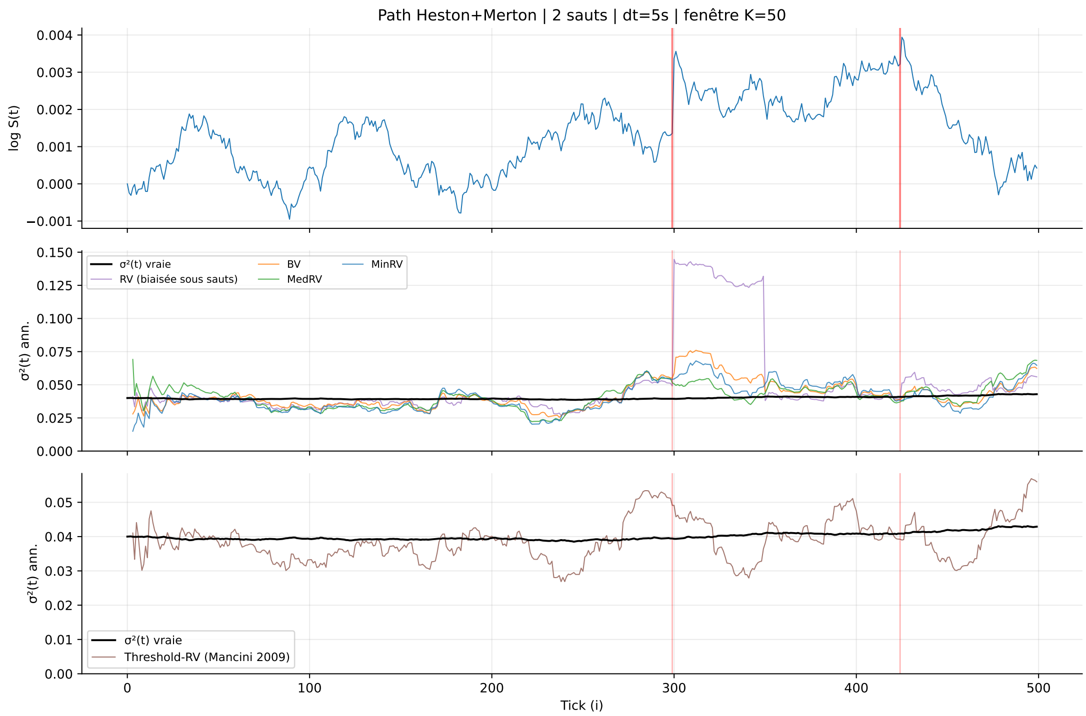


Figure 3: Comparison of spot volatility estimators (BV, MedRV, MinRV) on a simulated Heston path with Merton jumps. MedRV is the most robust to jump contamination.

Parameter `tau_factor` = 5. $\tau = 5\,\hat{\sigma}_i\sqrt{\Delta}$ places mass on jumps of size $\pm 5\,\hat{\sigma}_i\sqrt{\Delta}$, consistent with the grid regimes (3σ and 5σ).

5.3 Validity: $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] = 1$ under $H_{0,i}$

Proposition 5.1. *If $\hat{\sigma}_i$ is $\mathcal{F}_{i-1}$-measurable and $r_i \mid \mathcal{F}_{i-1} \sim \mathcal{N}(0, v)$ under $H_{0,i}$, then $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] = 1$.*

Proof. $E_i = \int (dP_J/dP_0)(r_i)\, d\pi(J)$ is the marginal density under the mixture divided by P_0. By Fubini: $\mathbb{E}_{P_0}[E_i] = \int \mathbb{E}_{P_0}[dP_J/dP_0]\, d\pi(J) = \int 1\, d\pi(J) = 1$. □

The measurability condition is ensured by excluding the tested point from the window (Section 4.3).

5.4 Empirical verification

Four unit tests verify:

1. **H0/H1 separation**: distribution of E_i concentrated near 1 under pure diffusion, shifted toward large values under a jump.
2. **Monotone e-power**: $\mathbb{E}[\log E_i]$ strictly increasing in jump size (from 3σ to 7σ).
3. **Dominance over p-to-e**: e-power of the GROW construction exceeds that of the $\kappa = 0.5$ calibrator.
4. **Validity**: $\mathbb{E}[E_i] \le 1 + 3\,\mathrm{SE}$ under H_0, $M = 10\,000$.

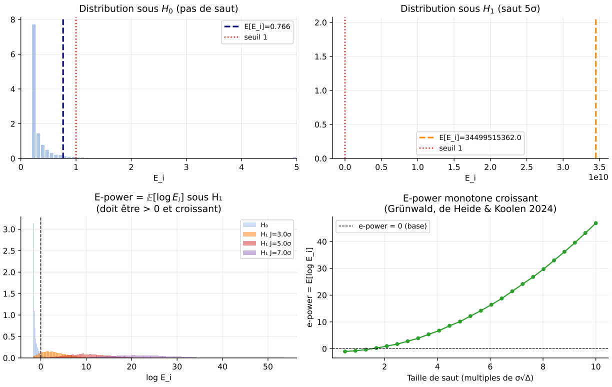
6 Online FDR control procedures

This section presents the seven procedures benchmarked in Section 8. The exposition follows the offline to online progression set up in Section 2.4: e-BH is offline, e-LOND, e-LORD, e-SAFFRON, and stopped e-BH are online. Proofs of FDR control rely on the e-value framework of Section 2.3.

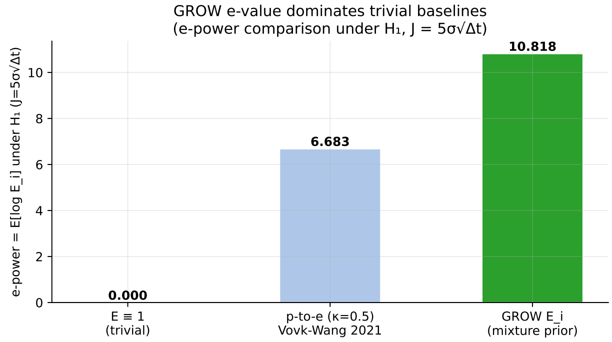
6.1 e-BH (Wang & Ramdas 2022)

The e-BH procedure is the e-value analogue of Benjamini-Hochberg. Before stating it, we need the notion of *compound e-variables*, which extends e-variables to sets of indices.

Definition 6.1 (Compound e-variables)**.** Let $H_1, \ldots, H_K$ be null hypotheses and $E_1, \ldots, E_K$



(a) Distribution of E_i under H_0 and H_1.



(b) E-power: GROW vs. Vovk-Wang calibrator ($\kappa = 0.5$).

Figure 4: Empirical validation of the GROW e-value construction. Left: under H_0 the distribution is concentrated near 1; under H_1 it is shifted to large values. Right: the GROW construction dominates the baseline calibrator in e-power across jump sizes.

non-negative random variables. They are *compound e-variables* for $(H_1, \dots, H_K)$ if

$$\mathbb{E}\left[\sum_{k \in N} E_k\right] \le |N|$$

for every configuration of true nulls $N \subseteq [K]$. They are *tight* if equality holds for some N.

A collection of K e-variables is automatically a collection of compound e-variables, but the converse is false in general.

Definition 6.2 (e-BH, Wang & Ramdas 2022)**.** Let $e_1, \dots, e_K$ be realised compound e-values and let $e_{[k]}$ denote the k-th order statistic from the largest to the smallest. The e-BH procedure at level $\alpha \in (0,1)$ rejects the hypotheses corresponding to the largest k^* e-values, where

$$k^* := \max\left\{k : \frac{k\, e_{[k]}}{K} \ge \frac{1}{\alpha}\right\}$$

($k^* := 0$ if the set is empty).

If $k^* = 0$, no hypothesis is rejected. The complexity is $O(K \log K)$ for sorting.

Theorem 6.3 (Wang & Ramdas 2022)**.** *For arbitrary dependence among the e-values, the e-BH procedure at level α satisfies*

$$\mathrm{FDR} \le \alpha \cdot \frac{|N|}{K} \le \alpha.$$

The proof factors through a generic property called *self-consistency*, which characterises a broad class of FDR-controlling procedures.

Definition 6.4 (Self-consistency)**.** An e-testing procedure D is *self-consistent at level α* if for every $k \in D$:

$$e_k \ge \frac{K}{\alpha\, R_D}.$$

Proposition 6.5. *The e-BH procedure is self-consistent. Moreover, any self-consistent e-testing procedure at level α has $\mathrm{FDR} \le \alpha K_0/K$ for arbitrary configurations of e-values, where $K_0 = |N|$.*

Proof. By self-consistency, for every $k \in D$: $\mathbb{1}_{\{k \in D\}}/R_D \le \alpha E_k/K$. Summing over N: $F_D/(R_D \vee 1) \le \sum_{k \in N} \alpha E_k/K$. Taking expectations and using $\mathbb{E}[E_k] \le 1$: $\mathrm{FDR} \le \alpha K_0/K$. Self-consistency of e-BH follows by construction: $e_k \ge e_{[R_D]} \ge K/(\alpha R_D)$. □

Theorem 6.3 follows by Proposition 6.5 applied to e-BH.

6.2 e-LOND (Xu & Ramdas 2024)

The online setting requires a different machinery: e-values arrive in a stream $(E_t)_{t \ge 1}$ and decisions must be made on the fly. We work with the natural filtration $\mathcal{F}_t = \sigma(E_1, \dots, E_t)$, $\mathcal{F}_0$ trivial.

Definition 6.6 (e-process)**.** An *e-process* is a non-negative adapted sequence $(E_t)_{t \ge 0}$ such that for every stopping time τ w.r.t. $(\mathcal{F}_t)$, $\mathbb{E}[E_\tau] \le 1$.

Since FDR is by definition the expectation of a proportion whose numerator and denominator may both be unbounded as $t \to \infty$, we work with the running quantity $\mathrm{FDR}(t)$.

Definition 6.7 (e-LOND, Xu & Ramdas 2024)**.** At step $t \geq 1$, set

$$\alpha_t = \alpha \cdot \gamma_t \cdot (R_{t-1} + 1),$$
$$\gamma_t = C_{\mathrm{JM}} \cdot \frac{\log \max(t, 2)}{t \cdot \exp\sqrt{\log \max(t, 2)}}, \quad (21)$$

where R_{t-1} is the number of rejections before time t and $(\gamma_t)_{t\geq 1}$ is a user-chosen non-increasing non-negative sequence satisfying $\sum_{t\geq 1} \gamma_t \leq 1$ (the choice above is the Javanmard-Montanari sequence). Reject H_t iff $E_t \geq 1/\alpha_t$.

Constant $C_{\mathrm{JM}} = 0.15708906$. Verified by truncated summation up to $t = 10^6$: $\sum_{t=1}^{10^6} \gamma_t \approx 1.00000$.

Interpretation. The budget $\alpha \sum_t \gamma_t \leq \alpha$ is distributed over time. The factor $(R_{t-1}+1)$ raises the rejection threshold after each rejection (wealth accumulation). The decay $\gamma_t \sim \log(t)/(t\, e^{\sqrt{\log t}})$ ensures validity for infinite streams.

Theorem 6.8 (Xu & Ramdas 2024)**.** *e-LOND controls* FDR $\leq \alpha$ *under arbitrary dependence.*

6.3 e-LORD (e-GAI framework, Zhang et al. 2025)

e-LORD takes a more flexible route via an *oracle FDP estimator*. The decision rule is again $\delta_t = \mathbb{1}_{\{E_t \geq 1/\alpha_t\}}$, but α_t is now built from an estimator that adapts to the rejection history.

Define the (unobservable) *oracle FDP*:

$$FDP^*(t) = \sum_{j \in H_0(t)} \frac{\alpha_j}{R_{j-1} + 1}.$$

It is called *oracle* because the summation runs over true nulls $H_0(t)$, which are unknown. The corresponding (observable) e-LORD estimator sums over all hypotheses:

$$\widehat{FDP}_{\text{e-LORD}}(t) = \sum_{j=1}^{t} \frac{\alpha_j}{R_{j-1} + 1}.$$

Proposition 6.9. *If* $\mathbb{E}[FDP^*(t)] \leq \alpha$ *for all* $t \geq 1$*, then* $\mathrm{FDR}(t) \leq \alpha$ *for all* $t \geq 1$.

Proof. By Markov $\delta_j \leq \alpha_j E_j$ and $R_t \vee 1 \geq R_{j-1} + 1$, so $\mathrm{FDR}(t) \leq \mathbb{E}[\sum_{j \in H_0} \alpha_j E_j/(R_{j-1} + 1)] \leq \mathbb{E}[\sum_{j \in H_0} \alpha_j/(R_{j-1} + 1)] = \mathbb{E}[FDP^*(t)] \leq \alpha$, using $\mathbb{E}[E_j \mid \mathcal{F}_{j-1}] \leq 1$. □

Since $H_0(t) \subseteq \{1, \ldots, t\}$ and all terms $\alpha_j/(R_{j-1} + 1) \geq 0$,

$$FDP^*(t) \leq \widehat{FDP}_{\text{e-LORD}}(t)$$

almost surely, giving the FDR guarantee.

Risk-aversion allocation rule. The testing levels α_t are set via the *Risk Aversion Investing* (RAI) rule of Zhang et al. (2025):

$$\alpha_t = \omega_t \cdot rw_t \cdot (R_{t-1} + 1),$$
$$rw_t = \alpha - \sum_{j=1}^{t-1} \frac{\alpha_j}{R_{j-1} + 1}, \quad (22)$$

where rw_t is the *remaining wealth* and $(\omega_t)_{t \in \mathbb{N}} \in (0, 1)$ governs the fraction of remaining wealth allocated to the current test. Initial wealth $rw_1 = \alpha$, $\alpha_1 = \alpha\omega_1$. The sequence ω_t updates as

$$\omega_{t+1} = \omega_t + \omega_1\, \phi^{t-R_t}(1 - \delta_t) - \omega_1\, \psi^{R_t} \delta_t, \quad (23)$$

with user-chosen parameters $\phi, \psi \in [0, 0.5]$ ensuring $\omega_t \in (0, 1)$.

Remark 6.10 (e-LOND as a special case). Setting $\gamma_t = \omega_t \prod_{j=1}^{t-1}(1 - \omega_j)$ in e-LOND recovers $\alpha_t^{\text{e-LOND}} = \alpha_t$ from (22). e-LOND is therefore a particular instantiation of the RAI framework with a static γ_t schedule.

Alpha-death. With $\omega_1 = 0.1$ over $n = 500$, e-LORD and e-SAFFRON exhaust their wealth by $t \approx 50$ (power ≈ 0.002). The canonical value is $\omega_1 = O(1/T)$ (Zhang et al. 2025): for $n = 500$, take $\omega_1 \leq 0.01$.

6.4 e-SAFFRON (e-GAI framework, Zhang et al. 2025)

e-SAFFRON is the *adaptive* variant of e-LORD. It introduces a predictable sequence $(\lambda_t)_{t\geq 1} \subset (0, 1)$ estimating the proportion of true nulls and *discards* candidate nulls whose e-values are too small, reallocating α-wealth toward more informative hypotheses.

The corresponding FDP estimator is

$$\widehat{FDP}_{\text{e-SAFFRON}}(t) = \sum_{j=1}^{t} \frac{\alpha_j}{R_{j-1} + 1} \cdot \frac{\mathbb{1}_{\{e_j < 1/\lambda_j\}}}{1 - \lambda_j}.$$

The intuition: when $e_j \geq 1/\lambda_j$, the hypothesis

Algorithm 1 e-LORD

Require: Initial wealth $rw_1 = \alpha$, sequence (ω_t), parameters ϕ, ψ

$R_0 \leftarrow 0$
for $t = 1, 2, \dots$ **do**
  $\alpha_t \leftarrow \omega_t \cdot rw_t \cdot (R_{t-1} + 1)$
  Observe e-value e_t
  if $e_t \geq 1/\alpha_t$ **then**
    Reject H_t; $\delta_t \leftarrow 1$
  else
    Accept H_t; $\delta_t \leftarrow 0$
  end if
  $R_t \leftarrow R_{t-1} + \delta_t$
  $rw_{t+1} \leftarrow rw_t - \alpha_t/(R_{t-1} + 1)$
  Update ω_{t+1} via (23)
end for

is a candidate alternative and is not charged to the FDP estimate; otherwise it contributes with inverse weight $(1 - \lambda_j)^{-1}$.

Theorem 6.11. *Let $(E_t)_{t \geq 1}$ be an e-process and $(\lambda_t)_{t \geq 1}$ a predictable sequence in $(0, 1)$. Then for every $t \geq 1$:*

$$\mathbb{E}[\widehat{FDP}_{e\text{-}SAFFRON}(t)] \geq \mathbb{E}[FDP^*(t)].$$

Proof. Since λ_j is $\mathcal{F}_{j-1}$-measurable, Markov gives $\mathbb{P}(E_j < 1/\lambda_j \mid \mathcal{F}_{j-1}) \geq 1 - \lambda_j$ under $H_{0,j}$. Restricting the sum to $j \in H_0(t)$ and applying this bound: $\mathbb{E}[\widehat{FDP}_{\text{e-SAFFRON}}(t)] \geq \sum_{j \in H_0(t)} \mathbb{E}[\alpha_j/(R_{j-1}+1)] = \mathbb{E}[FDP^*(t)]$. □

Combining with Proposition 6.9, e-SAFFRON controls FDR $\leq \alpha$ at every t.

Remaining-wealth rule. The testing levels use a modified wealth

$$rw_t^{\text{e-SAFFRON}} = \alpha(1 - \lambda) - \sum_{j=1}^{t-1} \frac{\alpha_j \, \mathbb{1}_{\{e_j < 1/\lambda_j\}}}{R_{j-1} + 1},$$

and $\alpha_t = \omega_t \cdot rw_t^{\text{e-SAFFRON}} \cdot (R_{t-1} + 1)$, with ω_t updated by (23). The rejection rule remains $\delta_t = \mathbb{1}_{\{e_t \geq 1/\alpha_t\}}$.

Setting $\lambda = 0$ recovers e-LORD: e-SAFFRON is therefore the adaptive counterpart of e-LORD.

6.5 Stopped e-BH (Wang, Dandapanthula & Ramdas 2025)

The offline e-BH procedure (Section 6.1) requires all e-values to be available at once. In sequential settings, data arrive over time and one may want to stop the experiment at a data-dependent time while still controlling FDR. The stopped e-BH procedure addresses this.

The key subtlety is the notion of filtration. Suppose G null hypotheses $H_0^1, \dots, H_0^G$ are tested simultaneously, each with its own *local* filtration $(\mathcal{F}_n^g)_{n \geq 0}$ generated from the data specific to hypothesis g. The *global* filtration is $\mathcal{F}_n = \sigma(\mathcal{F}_n^g : g \in [G])$. Let $\mathcal{T}$ denote the set of stopping times w.r.t. $(\mathcal{F}_n)$.

Definition 6.12 (Global e-process)**.** A non-negative adapted process $(M_n^g)_{n \geq 1}$ is a *global e-process* for $\mathcal{P}^g$ on $(\mathcal{F}_n)$ if at any stopping time $\tau \in \mathcal{T}$, M_τ^g is an e-value for $\mathcal{P}^g$: $\sup_{P \in \mathcal{P}^g} \mathbb{E}^P[M_\tau^g] \leq 1$.

A process that is an e-process only w.r.t. its local filtration $(\mathcal{F}_n^g)$ is a *local* e-process. The distinction is crucial: stopping a local e-process at a global stopping time $\tau \in \mathcal{T}$ does *not* in general yield a valid e-value, which would invalidate the FDR guarantee.

Definition 6.13 (Stopped FDR control)**.** An adapted sequence of random sets (R_n) with $R_n \subseteq [G]$ satisfies *level-α stopped FDR control* if

$$\sup_{P \in \mathcal{M}, \tau \in \mathcal{T}} FDR_\tau \leq \alpha,$$

where $FDR_\tau = \mathbb{E}^P[FDP_\tau]$ and $FDP_\tau = FD_\tau/(R_\tau \vee 1)$.

Definition 6.14 (Stopped e-BH)**.** Run e-BH at each time step on the current values of the e-processes: $R_n = eBH_\alpha(M_n^g : g \in [G])$.

Algorithm 2 e-SAFFRON

Require: Initial wealth $rw_1 = \alpha(1-\lambda)$, predictable sequence (λ_t), parameters ϕ, ψ

$R_0 \leftarrow 0$
for $t = 1, 2, \ldots$ **do**
  $\alpha_t \leftarrow \omega_t \cdot rw_t^{\text{e-SAFFRON}} \cdot (R_{t-1} + 1)$
  Observe e_t
  if $e_t \geq 1/\alpha_t$ **then**
    Reject H_t; $\delta_t \leftarrow 1$
  else
    Accept H_t; $\delta_t \leftarrow 0$
  end if
  $R_t \leftarrow R_{t-1} + \delta_t$
  $rw_{t+1}^{\text{e-SAFFRON}} \leftarrow rw_t^{\text{e-SAFFRON}} - \alpha_t \mathbb{1}_{\{e_t < 1/\lambda_t\}}/(R_{t-1} + 1)$
  Update ω_{t+1} via (23)
end for

Theorem 6.15 (Stopped e-BH, Wang et al. 2025)**.** *Let $(M_n^g)_{n\geq 1}$, $g \in [G]$, be global e-processes for $(\mathcal{P}^g : g \in [G])$. Then $(R_n) = \{eBH_\alpha(M_n^g : g \in [G])\}$ satisfies level-α stopped FDR control.*

Proof. Fix $\tau \in \mathcal{T}$ and $P \in \mathcal{M}$. Each (M_n^g) being a global e-process, the stopped values $(M_\tau^g : g \in [G])$ are valid e-values. Theorem 6.3 (FDR control of offline e-BH under arbitrary dependence) applied to these stopped e-values gives $FDR_\tau \leq \alpha$ for the set $eBH_\alpha(M_\tau^g : g \in [G])$. Since this holds for any τ and P, stopped FDR control follows. □

This reduces the problem to verifying that locally constructed e-processes are in fact global. Wang et al. (2025) identify a Markovian causal condition under which this holds.

Definition 6.16 (Markovian assumption, Wang et al. 2025, Assumption 3.1)**.** Consider observations $Z_n = (X_n, Y_n)$ where $X_n \in \mathcal{X}$ is a covariate and $Y_n = (Y_n^1, \ldots, Y_n^G) \in \mathcal{Y}^1 \times \cdots \times \mathcal{Y}^G$ a response vector. The Markovian assumption is

$$Y_n \perp (X_1, \ldots, X_{n-1}; Y_1, \ldots, Y_{n-1}) \mid X_n \quad \forall n \geq 2.$$

Given the current covariate X_n, Y_n is independent of the past, with arbitrary cross-component dependence in Y_n allowed.

Under this assumption, any e-process built multiplicatively from stepwise e-values computed within a single hypothesis is globally valid (Wang et al. 2025, Theorem 3.3), covering likelihood-ratio and universal-inference e-processes.

Application to jump detection. In our setting, the Markovian assumption is satisfied by the causal window of Section 4.3: $\hat{\sigma}_i$ is $\mathcal{F}_{i-1}$-measurable (does not look at r_i), so E_i is a global e-value. Complexity is $O(t \log t)$ per step.

6.6 Baselines BH-LM and BH-BNS

BH-LM. BH-LM computes $L_i = r_i/(\hat{\sigma}_i\sqrt{\Delta})$, the two-sided p-value $P_i = 2\mathbb{P}(|Z| > |L_i|)$, then applies BH. Decisions are made per hypothesis.

BH-BNS. BH-BNS applies a global BNS ratio test and duplicates its result across all hypotheses: if the global test rejects, all returns are flagged as jumps; otherwise none. This matches the BS 2016 usage (BNS as a day-level filter).

6.7 Synthetic comparison

7 Software architecture and pipeline

7.1 Five-layer structure

`simulate` $\to$ `estimators` $\to$ `evalues` $\to$ `fdr` $\to$ `pipeline`

7.2 Metrics

$$\text{FDP} = \tfrac{|\mathcal{R}\cap\mathcal{H}_0|}{\max(|\mathcal{R}|,1)}, \qquad \text{Power} = \tfrac{|\mathcal{R}\cap\mathcal{H}_1|}{\max(|\mathcal{H}_1|,1)},$$

$$\text{F1} = \tfrac{2\,\text{Prec}\cdot\text{Rec}}{\text{Prec}+\text{Rec}},$$

$$\text{Delay} = \tfrac{\sum_{i\in\mathcal{R}\cap\mathcal{H}_1}(t_i - t_{i,\text{true}})}{|\mathcal{R}\cap\mathcal{H}_1|}.$$

Table 2: Summary: guarantees, assumptions, complexity.

Procedure	Guarantee	Dependence	Key parameter	Complexity
e-BH	FDR $\leq \alpha$	Arbitrary	—	$O(n \log n)$
e-LOND	FDR $\leq \alpha$	Arbitrary	C_{JM}	$O(1)$/step
e-LORD	FDR $\leq \alpha$	Arbitrary	ω_1, ϕ, ψ	$O(1)$/step
e-SAFFRON	FDR $\leq \alpha$	Arbitrary	ω_1, λ	$O(1)$/step
Stopped e-BH	Anytime FDR $\leq \alpha$	Markov / causal	—	$O(t \log t)$/step
BH-LM	FDR $\leq \alpha$	PRDS	K_{win}	$O(n \log n)$
BH-BNS	FDR $\leq \alpha$	PRDS	—	$O(n)$

Reading the metrics. Each hypothesis $H_{0,i}$ targets a single tick i; the simulator provides the ground truth $\mathcal{H}_1$ (true jump indices). Three path-level quantities complement FDR and power:

- **Precision** $= |\mathcal{R} \cap \mathcal{H}_1| / \max(|\mathcal{R}|, 1)$: fraction of declared jumps that are genuine — the per-path analogue of $1 - \text{FDP}$.
- **Recall** $= |\mathcal{R} \cap \mathcal{H}_1| / \max(|\mathcal{H}_1|, 1)$: fraction of true jumps that are detected — the per-path analogue of power.
- **F1** $= 2\,\text{Precision}\cdot\text{Recall}/(\text{Precision}+\text{Recall})$: harmonic mean of the two; the natural scalar summary when precision and recall trade off against each other.
- **Delay**: average number of ticks between the true jump index and the rejection index, over correctly detected jumps. Since each hypothesis $H_{0,i}$ is decided at tick i (one-to-one correspondence), the delay is zero whenever the procedure rejects the exact tick of the jump and positive only if a neighbouring tick is rejected instead. Offline procedures (e-BH, BH-LM) are always zero-delay; online procedures may differ depending on the volatility window.

8 Empirical results

8.1 Protocol

Source: `results/grid_medium.parquet`

FDR violation convention. For $\alpha = 0.05$, $M = 100$: $\text{SE} = \sqrt{\alpha(1-\alpha)/M} = \sqrt{0.05 \times 0.95/100} \approx 0.0218$. Violation threshold: $\alpha + 2\,\text{SE} \approx 0.094$.

8.2 Result 1: BH's FDR violation at high frequency

Interpretation. The violation operates through two distinct channels. For BH-BNS, the BNS statistic is built on bipower variation, which is downward-biased whenever two consecutive returns both carry a jump — an event that becomes frequent under the moderate and hawkes_dense regimes at 5 s. The resulting under-estimation of integrated variance inflates the BNS ratio and triggers spurious detections. For BH-LM, the mechanism is more subtle: although MedRV is markedly more jump-robust than BV, the normalised statistics $L_i = r_i/(\hat{\sigma}_i\sqrt{\Delta})$ are computed over overlapping windows of $K = 50$ returns. At 5 s, these windows share a large fraction of their returns, and under dense jump clustering the dependence structure of the L_i violates the PRDS condition that BH requires to guarantee FDR $\leq \alpha$. In both cases the violation is structural: it does not disappear with larger sample size, only with a stronger theoretical guarantee (e.g. arbitrary-dependence e-value procedures).

Figure 5 illustrates the evolution of empirical FDR by frequency for $\alpha = 0.05$ and $\alpha = 0.10$.

8.3 Result 2: e-value-based FDR control at all frequencies

E-value-based procedures control the FDR at all tested frequencies (stopped e-BH ≤ 0.016, e-BH ≤ 0.010, e-LOND ≤ 0.001) under the three jump regimes, demonstrating robustness to the temporal dependence of Hawkes jumps.

Table 3: Parameters of the `grid_medium`.

Parameter	Value
n_{steps}	500
Frequencies (Δt, s)	[5, 60, 300]
Jump regimes	[rare, moderate, hawkes_dense]
Jump sizes (σ)	[3, 5]
α	[0.05, 0.10]
ω_1 (e-LORD/e-SAFFRON)	[0.1, 0.5]
M replications	100
Estimator	MedRV
Algorithms	7 (see Table 1)
Total tasks	32 400
Duration	$\approx$ 7 min

Table 4: Empirical FDR per algorithm $\times$ frequency ($\alpha = 0.05$, all regimes, $M = 100$, $n = 500$). Source: `grid_medium.parquet`.

Algorithm	$\Delta t = 5\,\mathrm{s}$	$\Delta t = 60\,\mathrm{s}$	$\Delta t = 300\,\mathrm{s}$	Violates?
BH-BNS	**0.096**	0.022	0.002	**Yes**
BH-LM	**0.096**	0.022	0.002	**Yes**
Stopped e-BH	0.016	0.005	0.000	No
e-BH	0.010	0.003	0.000	No
e-LOND	0.001	0.000	0.000	No
e-LORD ($\omega_1 = 0.1$)	0.000	0.000	0.000	No (alpha-death)
e-SAFFRON ($\omega_1 = 0.1$)	0.000	0.000	0.000	No (alpha-death)
Violation threshold: $0.094 = \alpha + 2\,\mathrm{SE}$				

8.4 Result 3: power-safety trade-off

Among procedures that actually control the FDR, **stopped e-BH** is the best compromise (power 0.349 vs. e-BH 0.326, e-LOND 0.235). The loss relative to BH (uncontrolled) is 9–21 pp at 5 s.

8.5 Result 4: FDR-power frontier

8.6 Result 5: Hawkes intensity and jump clustering

8.7 Result 6: detailed Hawkes dense regime

BH-BNS at 0.091 is below the violation threshold for hawkes_dense alone, but exceeds 0.094 once aggregated across all regimes (0.096). The temporal Hawkes dependence does not aggravate BH's violation — the violation is inherent to the 5 s frequency, not to clustering.

8.8 Result 7: effect of jump size

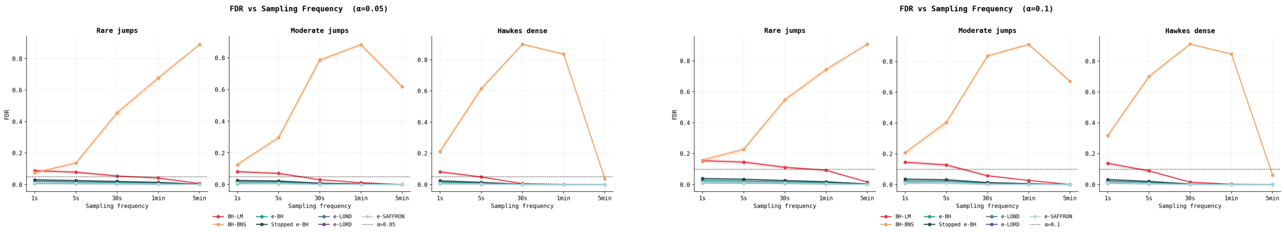
8.9 Result 8: heatmap view of aggregated performance

8.10 Result 9: BH violation and FDR — walkthrough views

8.11 Result 10: alpha-death and parameter ω_1

Alpha-death refers to the exhaustion of wealth $rw_t \to 0$ for e-LORD and e-SAFFRON. With $\omega_1 = 0.1$ over $n = 500$, both procedures yield power ≈ 0.002 (near zero): rw_t vanishes by $t \approx 50$.

The canonical value recommended by Zhang et al. 2025 is $\omega_1 = O(1/T)$. For $n = 500$, this corresponds to $\omega_1 \leq 0.01$. The results presented with $\omega_1 = 0.1$ illustrate the alpha-death phenomenon; properly calibrated values would restore power.



(a) $\alpha = 0.05$ (b) $\alpha = 0.10$

Figure 5: Empirical FDR vs. sampling frequency, by algorithm and regime (`grid_medium`, $M = 100$). The dashed line marks the nominal level α. BH-LM and BH-BNS exceed α at 5 s in all regimes.

Table 5: Power per algorithm $\times$ frequency ($\alpha = 0.05$, all regimes). Source: `grid_medium.parquet`.

Algorithm	$\Delta t = 5\,\mathrm{s}$	$\Delta t = 60\,\mathrm{s}$	$\Delta t = 300\,\mathrm{s}$
BH-BNS / BH-LM	0.441	0.358	0.157
Stopped e-BH	0.349	0.258	0.028
e-BH	0.326	0.252	0.027
e-LOND	0.235	0.168	0.013
e-LORD ($\omega_1 = 0.1$)	0.002	0.001	0.000
e-SAFFRON ($\omega_1 = 0.1$)	0.002	0.001	0.000

8.12 Result 11: computational performance

BH-LM is by far the fastest (simple normalisation). Stopped e-BH is the slowest ($O(T^2 \log T)$) but remains under 0.5 s for $n = 2000$.

9 Discussion and limitations

9.1 Debatable methodological choices

Symmetric Gaussian prior. The prior $\pi(J) = \mathcal{N}(0, \tau^2)$ favours no direction. If true jumps are systematically negative, an asymmetric prior would have higher e-power.

Window $K = 50$ not optimised. No sensitivity study on K: too small, the variance of $\hat{\sigma}_i$ is high; too large, it mixes different volatility regimes.

tau_factor $= 5.0$. A reasonable choice consistent with the grid jump sizes, but not derived from e-power optimisation.

SE convention. The convention $\mathrm{SE} = \sqrt{\alpha(1-\alpha)/M}$ (used here) is correct for testing $H_0 : \mathrm{FDR} \leq \alpha$. The empirical convention $\mathrm{SE} = \sqrt{\hat{p}(1-\hat{p})/M}$ gives a higher threshold (≈ 0.109) and would not flag a violation for BH at 0.096.

9.2 Validity of the results

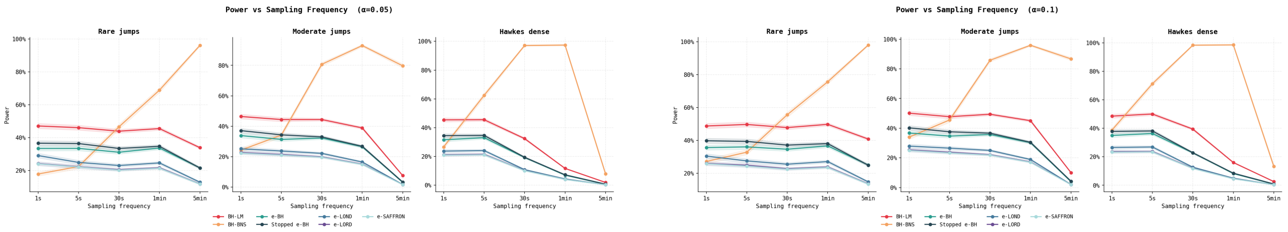
What is solid. E-value validity ($\mathbb{E}[E_i] \leq 1$ under H_0, 86 green tests), monotone e-power, dominance over p-to-e, and FDR control of e-value procedures at all frequencies are robustly established.

What is limited by $M = 100$. BH's FDR violation ($0.096 > 0.094$) is at the detection threshold. The 95% CI for BH-LM's FDR at 5 s is $[0.096 \pm 2 \times 0.022] = [0.052, 0.140]$, which includes the upper bound of the non-violation region. A larger M would confirm or refute the violation with greater statistical power.

10 Conclusion

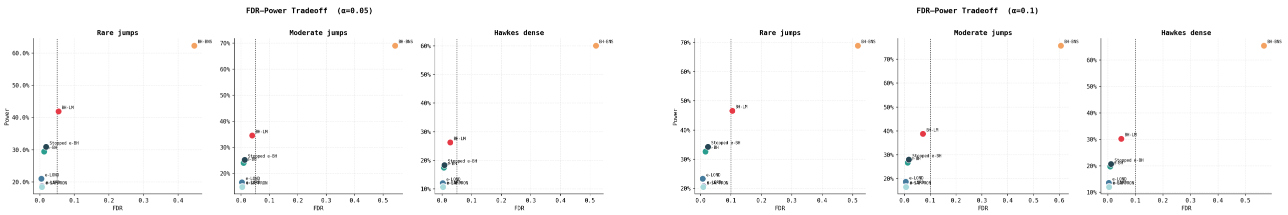
10.1 Synthesis

This project establishes that e-value-based FDR procedures control the false discovery rate at high frequency where classical methods (BH on Lee-Mykland p-values or BNS) fail. The mechanism is structural: BH requires PRDS, a condition



(a) $\alpha = 0.05$ (b) $\alpha = 0.10$

Figure 6: Power vs. sampling frequency by algorithm and regime. Stopped e-BH dominates e-BH and e-LOND at all frequencies among valid procedures. e-LORD/e-SAFFRON with $\omega_1 = 0.1$: alpha-death at 5 s.



(a) $\alpha = 0.05$ (b) $\alpha = 0.10$

Figure 7: Scatter of empirical FDR vs. power, by algorithm and regime. The dashed vertical line marks the nominal α. BH procedures sit to the right (uncontrolled FDR, high power); e-value procedures sit to the left (controlled FDR, lower power).

violated under dense jumps at 5 s, whereas e-values only require $\mathbb{E}[E_i \mid \mathcal{F}_{i-1}] \le 1$, valid under arbitrary dependence.

10.2 Quantified trade-off

At $\Delta t = 5$ s, $\alpha = 0.05$:

- BH: power 0.441, FDR ≈ 0.096.
- Stopped e-BH: power 0.349 (-21% relative), FDR ≤ 0.016.
- e-BH: power 0.326, FDR ≤ 0.010.
- e-LOND: power 0.235 (-47% relative), FDR ≤ 0.001.

The cost of guaranteed FDR control is 10–21 pp. Stopped e-BH offers the best power-safety compromise.

10.3 Perspectives

- Properly calibrate $\omega_1 = O(1/n)$ for e-LORD and e-SAFFRON, and re-evaluate their performance.
- Compare estimators BV, MinRV, threshold-BV, and MedRV in the grid.
- Apply to real intraday data (Bitcoin, TAQ, LOBSTER) to validate the pipeline outside of simulation.
- Multi-asset extension: co-jump detection (Jacod-Todorov 2009).

A Derivation of the GROW formula

Starting from (19), with $v = \Delta\hat{\sigma}^2$ and $\pi(J) = \mathcal{N}(0, \tau^2)$:

$$E_i = \int_{-\infty}^{+\infty} \frac{e^{Jr_i/v - J^2/(2v)}}{\sqrt{2\pi\tau^2}} \, e^{-J^2/(2\tau^2)} \, dJ.$$

Grouping terms in J:

$$-\frac{J^2}{2v} - \frac{J^2}{2\tau^2} + \frac{Jr_i}{v} = -\frac{(J-\mu^*)^2}{2s^2} + \frac{\mu^{*2}}{2s^2},$$

with $s^2 = v\tau^2/(v+\tau^2)$ and $\mu^* = r_i\tau^2/(v+\tau^2)$. The Gaussian integral gives s/τ, hence:

$$E_i = \tfrac{s}{\tau}\, e^{\mu^{*2}/(2s^2)} = \sqrt{\tfrac{v}{v+\tau^2}} \cdot e^{r_i^2\tau^2/[2v(v+\tau^2)]}.$$

This is exactly the expression implemented in `evalues/construct.py` (lines 40–49).

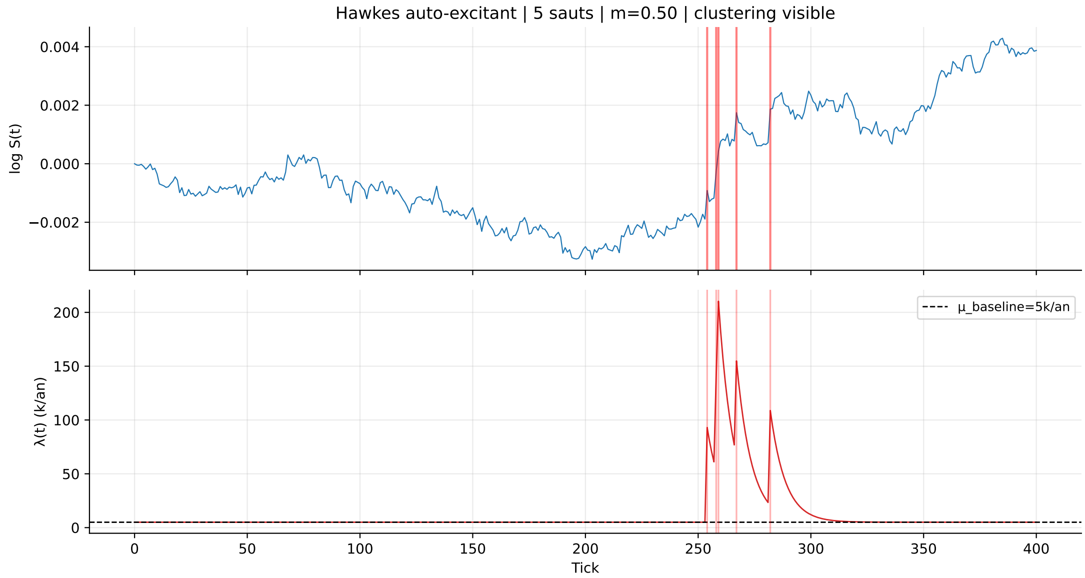


Figure 8: Simulated Hawkes intensity λ_t and corresponding jump arrivals on a single path ($\Delta t = 5\,\mathrm{s}$, hawkes_dense regime). Bursts of intensity reflect the self-exciting mechanism: each jump raises λ_t, generating temporal clustering.

Table 6: Detailed results: hawkes_dense, $\Delta t = 5\,\mathrm{s}$, $\alpha = 0.05$, $M = 100$.

Algorithm	Obs. FDR	Power	F1	FDR controlled?
BH-BNS	0.091	0.423	0.522	Borderline (< 0.094)
BH-LM	0.091	0.423	0.522	Borderline
Stopped e-BH	0.017	0.303	0.454	Yes
e-BH	0.009	0.280	0.435	Yes
e-LOND	0.000	0.204	0.371	Yes
e-LORD ($\omega_1 = 0.1$)	0.000	0.001	0.151	Yes (alpha-death)
e-SAFFRON ($\omega_1 = 0.1$)	0.000	0.001	0.151	Yes (alpha-death)

Table 7: Wall-clock per algorithm for $n = 2000$.

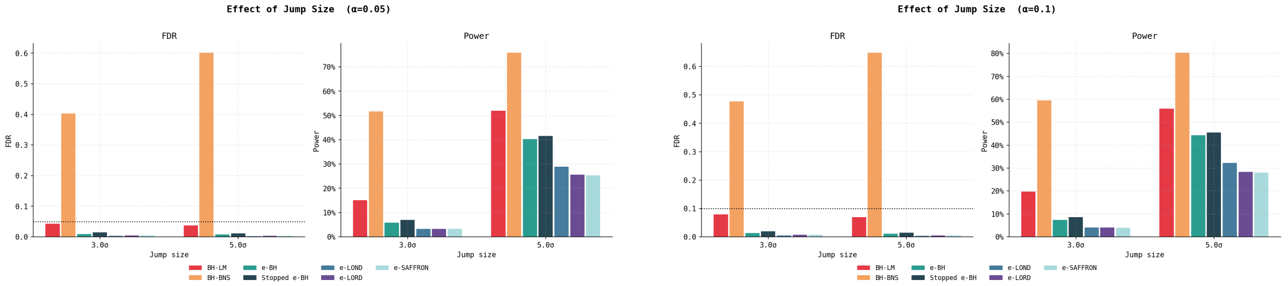
Algorithm	ms/run
BH-LM	1
e-BH	170
e-LOND	185
e-SAFFRON	174
e-LORD	187
BH-BNS	257
Stopped e-BH	346

B Javanmard-Montanari sequence for e-LOND

$$\gamma_t = C_{\mathrm{JM}} \cdot \frac{\log \max(t,2)}{t \cdot e^{\sqrt{\log \max(t,2)}}},$$

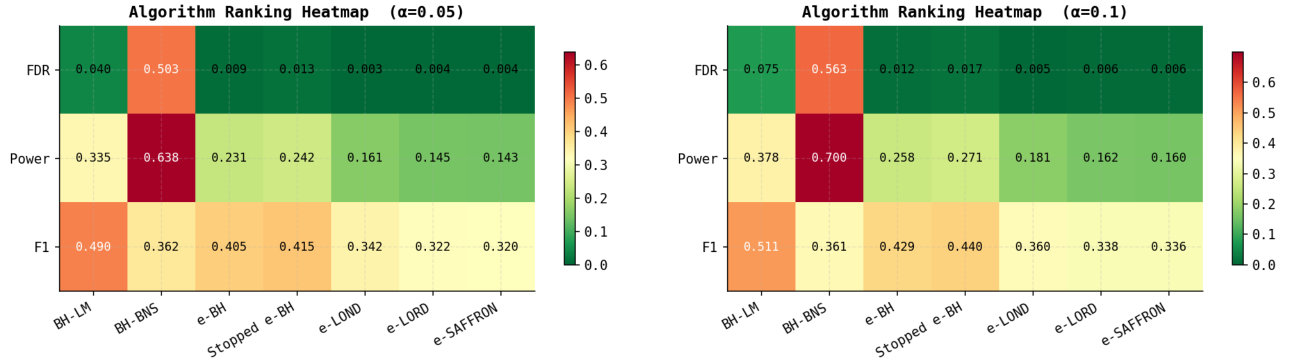
$$C_{\mathrm{JM}} = 0.15708906, \qquad \sum_{t=1}^{10^6} \gamma_t \approx 1.0000.$$

The normalisation $\sum_{t=1}^{\infty} \gamma_t = 1$ is the budget condition that ensures FDR control. C_{JM} is computed numerically by truncated summation (`fdr/elond.py` line 12).



(a) $\alpha = 0.05$ (b) $\alpha = 0.10$

Figure 9: FDR and power as a function of jump size (3σ vs. 5σ). Power increases with jump size for all procedures. BH's FDR remains above α irrespective of jump size at 5 s.



(a) $\alpha = 0.05$ (b) $\alpha = 0.10$

Figure 10: Heatmap of FDR, power, and F1 metrics by algorithm (aggregated across regimes and frequencies). Darker colours mark higher values. BH has the highest power but also the highest FDR. Stopped e-BH offers the best F1 among valid procedures.

C Commented code excerpts

C.1 Heston QE scheme (simulate/heston.py)

Listing 1: Case $\psi > 1.5$ (high dispersion) in the QE scheme.

```
p = (psi - 1.0) / (psi + 1.0)             # mass at 0
beta = 2.0 / (m * (psi + 1.0))            # exponential parameter
u = min(U[i], 1.0 - 1e-14)
if u <= p:
    vn = 0.0
else:
    vn = np.log((1.0 - p) / (1.0 - u)) / beta
```

C.2 Causal window (estimators/spot.py)

Listing 2: Window excluding the tested point i.

```
for i in range(n):
    start = max(0, i - K)
    sub = log_price[start : i + 1]
      # returns r[start:i] -- r[i] excluded
    if len(sub) < 4:
        continue
    sigma_sq[i] = estimator_fn(sub, dt_seconds)
```

C.3 E-value construction (evalues/construct.py)

Listing 3: GROW formula.

```
dv = dt_years * sigma_hat_i**2
tau = tau_factor * sigma_hat_i * np.sqrt(dt_years)
tau2 = tau**2
numerator = dv / (dv + tau2)
exponent  = r_i**2 * tau2 / (2 * dv * (dv + tau2))
```

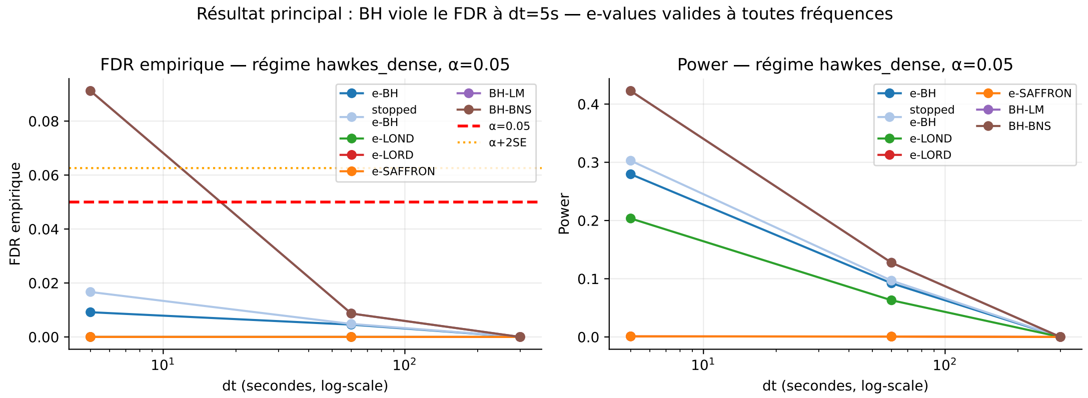
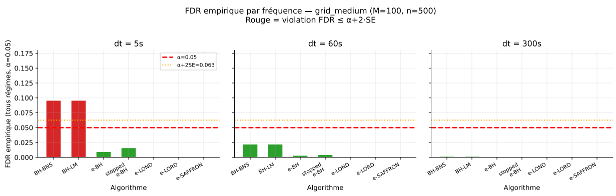
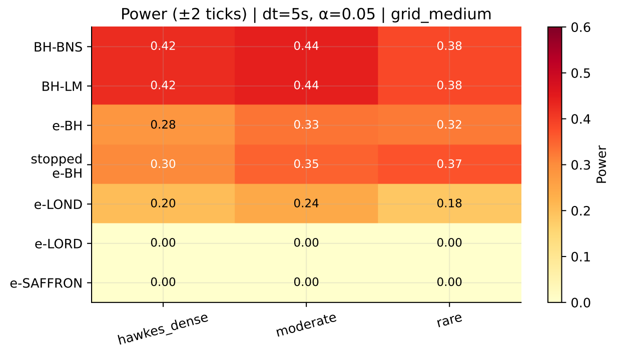


Figure 11: Zoom on the BH FDR violation at $\Delta t = 5\,\text{s}$: observed FDR per regime and jump size. The dashed line marks $\alpha + 2\,\text{SE} = 0.094$. BH-LM and BH-BNS exceed the threshold in the moderate and hawkes_dense regimes.



(a) FDR by frequency and procedure.



(b) Power heatmap by regime and frequency.

Figure 12: Additional walkthrough views from `00_project_walkthrough.ipynb`, complementing Figures 5 and 10.

```
e_value    = np.sqrt(numerator) * np.
    exp(exponent)
```

C.4 e-LOND (`fdr/elond.py`)

Listing 4: Main loop.

```
R = 0
for t, e_t in enumerate(e_values,
    start=1):
     alpha_t = alpha * _gamma(t) * (R
          + 1)
     reject = e_t >= 1.0 / alpha_t if
          alpha_t > 0 else False
     if reject:
          R += 1
     yield reject
```

C.5 Stopped e-BH (`fdr/stopped_ebh.py`)

Listing 5: Offline e-BH recomputed at each step.

```
buffer: list[float] = []
for e_t in e_values:
    buffer.append(e_t)
    t = len(buffer)
    ev = np.asarray(buffer, dtype=
        float)
    order = np.argsort(ev)[::-1]
    thresholds = t / (alpha * np.
        arange(1, t + 1))
    qualifying = np.where(ev[order]
        >= thresholds)[0]
    if len(qualifying) == 0:
        yield False
    else:
        k_star = int(qualifying[-1])
            + 1
        yield bool(e_t >= t / (alpha
            * k_star))
```

D Glossary

Alpha-death Exhaustion of the wealth $rw_t \to 0$ for e-LORD/e-SAFFRON when ω_1 is too large relative to T.

Anytime-validity FDR controlled at every random stopping time τ.

Branching ratio $m = \alpha_H/(\beta_H \times T)$ for a Hawkes process; $m < 1$ implies stationarity.

E-power $\mathbb{E}_{H_1}[\log E]$: criterion of sequential efficiency.

E-variable $E \geq 0$ with $\mathbb{E}[E \mid H_0] \leq 1$.

FDR False Discovery Rate: $\mathbb{E}[|\mathcal{R} \cap \mathcal{H}_0| / \max(|\mathcal{R}|, 1)]$.

FDP False Discovery Proportion: random realisation of the FDR.

GROW Growth-Rate Optimal in the Worst case: e-value maximising e-power under $\mathbb{E}_{H_0}[E] \leq 1$.

PRDS Positive Regression Dependence on a Subset: sufficient dependence condition for the validity of BH.

RAI Risk Aversion Investing: framework of e-LORD/e-SAFFRON.

Self-consistency Property characterising a broad class of FDR-controlling procedures, generalising e-BH.

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